

# Tutorial: Carrier Diffusion and Non-Equilibrium Recombination

(Direct/Radiative, Trap-Assisted SRH, and Auger)

Solid State Physics Lessons

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## Useful data (unless otherwise stated)

- Elementary charge  $q = 1.602 \times 10^{-19}$  C,  $kT/q \approx 0.0259$  V at 300 K
- Silicon (Si, 300 K):  $n_i \approx 1.0 \times 10^{10} \text{ cm}^{-3}$ , typical mobilities  $\mu_n = 1350 \text{ cm}^2/(\text{V s})$ ,  $\mu_p = 480 \text{ cm}^2/(\text{V s})$ .
- Einstein relation (non-degenerate, near equilibrium):  $D = \mu kT/q$ .
- Radiative recombination (direct-gap):  $R_{\text{rad}} = B(np - n_i^2)$ ,  $B$  e.g. GaAs  $\sim 1 \times 10^{-10} \text{ cm}^3/\text{s}$ ; Si is much smaller ( $\sim 10^{-14}$ – $10^{-15}$ ).
- SRH recombination:

$$U_{\text{SRH}} = \frac{np - n_i^2}{\tau_{p0}(n + n_1) + \tau_{n0}(p + p_1)}, \quad n_1 = n_i e^{(E_t - E_i)/kT}, \quad p_1 = n_i e^{-(E_t - E_i)/kT}.$$

- Auger recombination:

$$R_{\text{Aug}} = C_n n(np - n_i^2) + C_p p(np - n_i^2)$$

with typical Si values  $C_n \sim 2.8 \times 10^{-31}$  and  $C_p \sim 1.0 \times 10^{-31} \text{ cm}^6/\text{s}$ .

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## 1 Part A — Numerical Problems (10)

### Problem 1: Mixed drift–diffusion currents (Si)

**Problem Statement.** Given  $\mu_n = 1350 \text{ cm}^2/(\text{V s})$ ,  $\mu_p = 480 \text{ cm}^2/(\text{V s})$ ,  $E = 20 \text{ V/cm}$ ,  $n = 5.0 \times 10^{15} \text{ cm}^{-3}$ ,  $p = 1.0 \times 10^{14} \text{ cm}^{-3}$ ,  $\frac{dn}{dx} = -5.0 \times 10^{17} \text{ cm}^{-4}$ ,  $\frac{dp}{dx} = 2.0 \times 10^{17} \text{ cm}^{-4}$ . Find  $J_n$  and  $J_p$  ( $\text{A/cm}^2$ ), and identify drift/diffusion parts.

**Solution.**

*Step 1: Calculate diffusion coefficients using Einstein relation.*

At thermal equilibrium (or under weak perturbation), the Einstein relation connects mobility to diffusivity:

$$D_n = \mu_n \frac{kT}{q} = 1350 \times 0.0259 = 34.96 \text{ cm}^2/\text{s} \approx 35.0 \text{ cm}^2/\text{s}.$$

$$D_p = \mu_p \frac{kT}{q} = 480 \times 0.0259 = 12.43 \text{ cm}^2/\text{s} \approx 12.4 \text{ cm}^2/\text{s}.$$

*Step 2: Write drift-diffusion current equations.*

The total current has two components: drift (due to electric field) and diffusion (due to concentration gradient):

$$J_n = q \left( n\mu_n E + D_n \frac{dn}{dx} \right), \quad J_p = q \left( p\mu_p E - D_p \frac{dp}{dx} \right).$$

Note the minus sign for holes: it accounts for the opposite charge sign.

*Step 3: Calculate electron drift current.*

Drift occurs when the electric field  $E > 0$  (pointing in  $+x$ ) accelerates carriers. For electrons:

$$\begin{aligned} J_{n,\text{drift}} &= qn\mu_n E = (1.602 \times 10^{-19}) \times (5.0 \times 10^{15}) \times 1350 \times 20. \\ &= 21.63 \text{ A/cm}^2. \end{aligned}$$

*Step 4: Calculate electron diffusion current.*

Since  $dn/dx = -5.0 \times 10^{17} < 0$ , electron concentration decreases in the  $+x$  direction. Electrons diffuse toward  $+x$  (down gradient), carrying negative charge, so the conventional current flows in  $-x$ :

$$\begin{aligned} J_{n,\text{diff}} &= qD_n \frac{dn}{dx} = (1.602 \times 10^{-19}) \times 35.0 \times (-5.0 \times 10^{17}). \\ &= -2.80 \text{ A/cm}^2. \end{aligned}$$

*Step 5: Calculate total electron current.*

$$J_n = J_{n,\text{drift}} + J_{n,\text{diff}} = 21.63 + (-2.80) = \boxed{18.83 \text{ A/cm}^2}.$$

The net flow is in the  $+x$  direction, dominated by drift.

*Step 6: Calculate hole currents.*

Hole drift:

$$J_{p,\text{drift}} = qp\mu_p E = (1.602 \times 10^{-19}) \times (1.0 \times 10^{14}) \times 480 \times 20 = 0.154 \text{ A/cm}^2.$$

Hole diffusion (note the minus sign):

$$J_{p,\text{diff}} = -qD_p \frac{dp}{dx} = -(1.602 \times 10^{-19}) \times 12.4 \times (2.0 \times 10^{17}) = -0.398 \text{ A/cm}^2.$$

Total hole current:

$$J_p = 0.154 + (-0.398) = \boxed{-0.244 \text{ A/cm}^2}.$$

Holes flow in the  $-x$  direction, driven by the strong diffusion gradient.

**Physical interpretation:** The electron current is dominated by drift (carried by the large majority concentration  $n \gg p$ ), while hole current is dominated by diffusion. Current continuity requires  $J_n + J_p = \text{constant}$  (the total current).

## Problem 2: Built-in field in a graded $n$ -type bar

**Problem Statement.** An  $n$ -type Si bar has  $n(x) = N_0 e^{ax}$  with  $N_0 = 1 \times 10^{16} \text{ cm}^{-3}$ ,  $a = 300 \text{ cm}^{-1}$  at 300 K. Find the electric field  $E(x)$  such that  $J_n = 0$  (no-current condition).

**Solution.**

*Step 1: Apply the zero-current condition.*

To have no electron current everywhere, drift and diffusion must exactly cancel:

$$0 = J_n = q(n\mu_n E + D_n dn/dx).$$

*Step 2: Solve for the electric field.*

Rearranging:

$$E = -\frac{D_n}{\mu_n} \frac{1}{n} \frac{dn}{dx}.$$

Using the Einstein relation  $D_n/\mu_n = kT/q$ :

$$E(x) = -\frac{kT}{q} \frac{1}{n(x)} \frac{dn}{dx}.$$

*Step 3: Calculate the logarithmic derivative.*

For  $n(x) = N_0 e^{ax}$ :

$$\frac{d}{dx} \ln n(x) = \frac{1}{n} \frac{dn}{dx} = \frac{d}{dx}(ax) = a.$$

*Step 4: Compute the built-in field.*

$$E(x) = -\frac{kT}{q} \cdot a = -0.0259 \times 300 = \boxed{-7.77 \text{ V/cm}}.$$

**Key result:** The field is *uniform* (independent of  $x$ ) even though the doping varies exponentially! This is the equilibrium condition in a graded material: the built-in field exactly compensates for diffusion, maintaining zero current.

## Problem 3: Diffusion length from $D$ and $\tau$

**Problem Statement.** Minority electrons in  $p$ -type Si have  $D_n = 30 \text{ cm}^2/\text{s}$  and lifetime  $\tau_n = 5.0 \mu\text{s}$ . Find the diffusion length  $L_n = \sqrt{D_n \tau_n}$ .

**Solution.**

*Step 1: Understand the physical significance.*

The diffusion length  $L_n$  characterizes how far minority electrons can travel by random diffusion before recombining. It is the characteristic length scale for minority carrier transport in the quasi-neutral base region of bipolar transistors and the collection region of photodiodes.

*Step 2: Calculate the diffusion length.*

$$\begin{aligned} L_n &= \sqrt{D_n \tau_n} = \sqrt{30 \times 5.0 \times 10^{-6}} = \sqrt{1.5 \times 10^{-4}} \\ &= \boxed{0.0122 \text{ cm} = 122 \mu\text{m}}. \end{aligned}$$

**Physical context:** - If the  $p$ -type base width  $< L_n$ , minority electrons traverse it quickly and are collected at the other contact - If base width  $\gg L_n$ , most electrons recombine before escaping - For efficient transistor operation, we need base width  $\sim 1-10 \times L_n$

## Problem 4: Turn-off transient

**Problem Statement.** Uniform excess carrier density  $\Delta n(0) = 1.0 \times 10^{13} \text{ cm}^{-3}$  exists in  $p$ -Si. The bulk lifetime is  $\tau = 4.0 \mu\text{s}$ . Find (i)  $\Delta n(t)$  and (ii) the time for excess to decay to 10% of its initial value.

**Solution.**

*Step 1: Understand recombination kinetics.*

When excess carriers are present in a bulk semiconductor far from surfaces, they recombine at a first-order rate proportional to the excess concentration:

$$\frac{d(\Delta n)}{dt} = -\frac{\Delta n}{\tau}.$$

*Step 2: Solve the differential equation.*

This is a standard exponential decay:

$$\Delta n(t) = \Delta n(0) e^{-t/\tau}.$$

*Step 3: Find the time to reach 10%.*

Set  $\Delta n(t) = 0.1 \times \Delta n(0)$ :

$$0.1 = e^{-t/\tau} \implies -\ln(0.1) = t/\tau \implies t = \tau \ln(10).$$

$$t = 4.0 \mu\text{s} \times 2.303 = \boxed{9.21 \mu\text{s}}.$$

**Practical relevance:** This decay is observed experimentally via transient photoconductivity (TPV) measurements. By switching off illumination and observing the voltage decay, researchers extract lifetime values directly.

## Problem 5: Steady-state photoconductivity

**Problem Statement.** Uniform generation  $g = 3.0 \times 10^{19} \text{ cm}^{-3}\text{s}^{-1}$  is sustained under illumination. Low-level injection;  $\tau = 1.0 \mu\text{s}$ ;  $\mu_n = 1350 \text{ cm}^2/(\text{V s})$ ,  $\mu_p = 480 \text{ cm}^2/(\text{V s})$ . Find  $\Delta n = \Delta p$  and the photoconductivity change  $\Delta\sigma$ .

**Solution.**

*Step 1: Apply steady-state balance.*

Under continuous illumination, generation is balanced by recombination:

$$g = \frac{\Delta n}{\tau}.$$

*Step 2: Calculate excess carrier density.*

$$\Delta n = g\tau = 3.0 \times 10^{19} \times 1.0 \times 10^{-6} = \boxed{3.0 \times 10^{13} \text{ cm}^{-3}}.$$

By charge neutrality in the quasi-neutral region,  $\Delta p = \Delta n$ .

*Step 3: Calculate photoconductivity increase.*

Both electrons and holes contribute to conductivity:

$$\begin{aligned} \Delta\sigma &= q\Delta n(\mu_n + \mu_p) = q\Delta n(\mu_n + \mu_p). \\ &= (1.602 \times 10^{-19}) \times (3.0 \times 10^{13}) \times (1350 + 480). \end{aligned}$$

$$= (1.602 \times 10^{-19}) \times (3.0 \times 10^{13}) \times 1830.$$

$$= \boxed{8.79 \times 10^{-3} \Omega^{-1} \text{cm}^{-1}}.$$

**Physical insight:** The conductivity change is proportional to both the generation rate and the lifetime. Longer lifetimes or higher generation rates lead to larger photoconductivity the basis for photodetectors and solar cells.

## Problem 6: SRH recombination rate in $n$ -type Si (midgap trap)

**Problem Statement.**  $n$ -type Si ( $n_0 = 1.0 \times 10^{16} \text{ cm}^{-3}$ ,  $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$ ) with a trap at midgap ( $E_t = E_i$ , so  $n_1 = p_1 = n_i$ ). SRH lifetimes:  $\tau_{p0} = 10 \mu\text{s}$ ,  $\tau_{n0} = 20 \mu\text{s}$ . Uniform excess  $\Delta n = \Delta p = 5.0 \times 10^{13} \text{ cm}^{-3}$  (low-level). Find  $U_{\text{SRH}}$  and effective lifetime  $\tau_{\text{eff}} = \Delta n / U_{\text{SRH}}$ .

**Solution.**

*Step 1: Determine total carrier concentrations.*

Electrons:  $n = n_0 + \Delta n = 1.0 \times 10^{16} + 5.0 \times 10^{13} \approx 1.005 \times 10^{16} \text{ cm}^{-3}$ .

Holes: First find equilibrium:  $p_0 = n_i^2 / n_0 = (10^{10})^2 / 10^{16} = 1.0 \times 10^4 \text{ cm}^{-3}$ . Then  $p = p_0 + \Delta p \approx 5.0 \times 10^{13} \text{ cm}^{-3}$  (excess dominates).

*Step 2: Calculate the mass action product deviation.*

$$np - n_i^2 = (1.005 \times 10^{16})(5.0 \times 10^{13}) - (10^{10})^2 = 5.025 \times 10^{29} \text{ cm}^{-6}.$$

Since  $n_i^2 = 10^{20}$  is negligible,  $np \approx 5.025 \times 10^{29} \text{ cm}^{-6}$ .

*Step 3: Calculate the SRH denominator.*

The denominator combines capture time constants weighted by carrier densities:

$$\tau_{p0}(n + n_1) + \tau_{n0}(p + p_1).$$

Substituting:  $n_1 = p_1 = n_i = 10^{10}$  (midgap trap):

$$= 10 \times 10^{-6} \times (1.005 \times 10^{16} + 10^{10}) + 20 \times 10^{-6} \times (5.0 \times 10^{13} + 10^{10}).$$

$$\approx 10^{-5} \times 1.005 \times 10^{16} + 2 \times 10^{-5} \times 5.0 \times 10^{13}.$$

$$= 1.005 \times 10^{11} + 1.0 \times 10^9 \approx 1.015 \times 10^{11} \text{ s}.$$

(The first term dominates because  $n \gg p$  in  $n$ -type material.)

*Step 4: Calculate SRH recombination rate.*

$$U_{\text{SRH}} = \frac{np - n_i^2}{\tau_{p0}(n + n_1) + \tau_{n0}(p + p_1)} = \frac{5.025 \times 10^{29}}{1.015 \times 10^{11}}.$$

$$= \boxed{4.95 \times 10^{18} \text{ cm}^{-3} \text{s}^{-1}}.$$

*Step 5: Calculate effective lifetime.*

The effective lifetime is the ratio of excess to recombination rate:

$$\tau_{\text{eff}} = \frac{\Delta n}{U_{\text{SRH}}} = \frac{5.0 \times 10^{13}}{4.95 \times 10^{18}} = \boxed{1.01 \times 10^{-5} \text{ s}} \approx \boxed{10.1 \mu\text{s}}.$$

**Physical interpretation:** - Under low-level injection in  $n$ -type Si, the SRH lifetime is relatively long ( $\sim 10 \text{ s}$ ) - This is because the denominator is dominated by  $\tau_{p0}n$  (capturing electrons with slow hole capture) - The minority hole population ( $p \approx 10^{13} \text{ cm}^{-3}$ ) is too small to quickly capture electrons - This is why deep traps (like midgap) give longer lifetimes in heavily doped  $n$ -type material

## Problem 7: Radiative recombination lifetime in GaAs (high-level injection)

**Problem Statement.** GaAs at 300 K with radiative coefficient  $B = 1.0 \times 10^{-10} \text{ cm}^3/\text{s}$ . High-level injection:  $\Delta n = \Delta p = 1.0 \times 10^{16} \text{ cm}^{-3}$  in intrinsic material (assume  $n_i \ll \Delta n$ ). Find  $R_{\text{rad}}$  and  $\tau_{\text{rad}} = \Delta n / R_{\text{rad}}$ .

**Solution.**

*Step 1: Understand direct recombination.*

In direct-gap semiconductors (GaAs, InP), an electron from the conduction band can directly recombine with a hole in the valence band, emitting a photon. The rate is:

$$R_{\text{rad}} = B(np - n_i^2).$$

*Step 2: Calculate carrier product under high-level injection.*

When  $\Delta n = \Delta p$  and both are much larger than intrinsic carrier concentration:

$$n = n_i + \Delta n \approx \Delta n = 1.0 \times 10^{16}, \quad p = p_i + \Delta p \approx \Delta n = 1.0 \times 10^{16}.$$

Thus:

$$np - n_i^2 \approx (\Delta n)^2 = (1.0 \times 10^{16})^2 = 1.0 \times 10^{32} \text{ cm}^{-6}.$$

*Step 3: Calculate radiative recombination rate.*

$$R_{\text{rad}} = B \times (\Delta n)^2 = 1.0 \times 10^{-10} \times 1.0 \times 10^{32} = \boxed{1.0 \times 10^{22} \text{ cm}^{-3}\text{s}^{-1}}.$$

*Step 4: Calculate radiative lifetime.*

$$\tau_{\text{rad}} = \frac{\Delta n}{R_{\text{rad}}} = \frac{1.0 \times 10^{16}}{1.0 \times 10^{22}} = \boxed{1.0 \times 10^{-6} \text{ s} = 1.0 \mu\text{s}}.$$

**Physical significance:** - GaAs has a direct bandgap, making radiative recombination efficient - At high injection levels, radiative lifetime becomes the limiting factor - This is the basis for LEDs and laser diodes: injected carriers recombine radiatively - Compare to silicon (indirect gap): radiative lifetime would be  $\sim 100 \text{ s}$  at the same injection (much slower)

## Problem 8: Auger-limited lifetime in Si (high-level injection)

**Problem Statement.** Si at 300 K with Auger coefficients  $C_n = 2.8 \times 10^{-31} \text{ cm}^6/\text{s}$ ,  $C_p = 1.0 \times 10^{-31} \text{ cm}^6/\text{s}$ . Quasi-intrinsic (equal carrier injection):  $n = p = \Delta n = 5.0 \times 10^{17} \text{ cm}^{-3}$ . Find  $R_{\text{Aug}}$  and  $\tau_{\text{Aug}} = \Delta n / R_{\text{Aug}}$ .

**Solution.**

*Step 1: Understand Auger recombination.*

In Auger processes, an electron-hole pair recombines, transferring energy to a third carrier (electron or hole) which becomes heated. This nonradiative process dominates at high injection in Si:

$$R_{\text{Aug}} = (C_n n + C_p p)(np - n_i^2).$$

*Step 2: Simplify under quasi-intrinsic conditions.*

When  $n = p = \Delta n \gg n_i$ :

$$np - n_i^2 \approx n^2 = p^2 = (\Delta n)^2.$$

Thus:

$$R_{\text{Aug}} = (C_n n + C_p p)(\Delta n)^2 = (C_n + C_p)\Delta n^3.$$

Step 3: Calculate  $(\Delta n)^3$ .

$$(\Delta n)^3 = (5.0 \times 10^{17})^3 = 125 \times 10^{51} = 1.25 \times 10^{53} \text{ cm}^{-9}.$$

Step 4: Calculate Auger recombination rate.

$$\begin{aligned} R_{\text{Aug}} &= (C_n + C_p)\Delta n^3 = (2.8 + 1.0) \times 10^{-31} \times 1.25 \times 10^{53}. \\ &= 3.8 \times 10^{-31} \times 1.25 \times 10^{53} = \boxed{4.75 \times 10^{22} \text{ cm}^{-3} \text{ s}^{-1}}. \end{aligned}$$

Step 5: Calculate Auger lifetime.

$$\tau_{\text{Aug}} = \frac{\Delta n}{R_{\text{Aug}}} = \frac{5.0 \times 10^{17}}{4.75 \times 10^{22}} = \boxed{1.05 \times 10^{-5} \text{ s}} \approx \boxed{10.5 \mu\text{s}}.$$

**Key observation:** - Auger lifetime depends on  $\Delta n^3$  (cubic!), not  $\Delta n$  like other mechanisms - At moderate injection ( $10^{15} \text{ cm}^{-3}$ ):  $\tau_{\text{Aug}} \gg \tau_{\text{SRH}}$  (slow) - At high injection ( $10^{17} \text{ cm}^{-3}$ ):  $\tau_{\text{Aug}}$  becomes competitive with radiative (in GaAs) or SRH (in Si) - This cubic dependence is why high-efficiency solar cells must carefully manage injection levels

## Problem 9: Effective lifetime with surface recombination

**Problem Statement.** Si slab thickness  $W = 200 \mu\text{m}$ , ambipolar diffusion coefficient  $D = 15 \text{ cm}^2/\text{s}$ , bulk lifetime  $\tau_b = 30 \mu\text{s}$ . Both surfaces act as strong recombination sinks. Using first-mode approximation, find  $\tau_{\text{eff}}$ :

$$\frac{1}{\tau_{\text{eff}}} = \frac{1}{\tau_b} + \frac{\pi^2 D}{W^2}.$$

**Solution.**

Step 1: Understand the two recombination channels.

Carriers can recombine via:

1. **Bulk recombination:** Spontaneous (SRH, radiative, Auger) throughout the volume lifetime  $\tau_b$
2. **Surface recombination:** Diffusion to high-recombination boundaries timescale  $\sim W^2/D$

These are independent processes; rates add:  $1/\tau_{\text{eff}} = 1/\tau_b + 1/\tau_{\text{surface}}$ .

Step 2: Convert units.

$$W = 200 \mu\text{m} = 200 \times 10^{-4} \text{ cm} = 2.0 \times 10^{-2} \text{ cm}.$$

Step 3: Calculate the bulk recombination rate.

$$\frac{1}{\tau_b} = \frac{1}{30 \times 10^{-6}} = 3.33 \times 10^4 \text{ s}^{-1}.$$

Step 4: Calculate surface recombination contribution (diffusion-limited).

The rate at which carriers reach the surface by diffusion is characterized by:

$$\begin{aligned}\frac{\pi^2 D}{W^2} &= \frac{\pi^2 \times 15}{(2.0 \times 10^{-2})^2} = \frac{9.8696 \times 15}{4.0 \times 10^{-4}} = \frac{148.04}{4.0 \times 10^{-4}} \\ &= 3.701 \times 10^5 \text{ s}^{-1}.\end{aligned}$$

*Step 5: Calculate effective lifetime.*

$$\begin{aligned}\frac{1}{\tau_{\text{eff}}} &= 3.33 \times 10^4 + 3.701 \times 10^5 = 4.034 \times 10^5 \text{ s}^{-1} \\ \tau_{\text{eff}} &= \frac{1}{4.034 \times 10^5} = \boxed{2.48 \mu\text{s}}.\end{aligned}$$

**Physical insight:** - Surface recombination dominates by a factor  $\sim 11$  - Effective lifetime is much shorter than bulk (2.48 vs 30 s) - This demonstrates why surface passivation is critical for silicon devices - With good passivation (oxide layer): surface recombination velocity drops from  $10^5$  to 1 cm/s, potentially extending  $\tau_{\text{eff}}$  to  $\sim 20$  s

## Problem 10: Ambipolar diffusion coefficient and transit time

**Problem Statement.**  $n$ -type Si:  $n_0 = 5.0 \times 10^{15} \text{ cm}^{-3}$ ,  $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$ ,  $\mu_n = 1200 \text{ cm}^2/(\text{V s})$ ,  $\mu_p = 450 \text{ cm}^2/(\text{V s})$ . Low-level optical excitation (minority holes dominate). Find  $D_n$ ,  $D_p$ ,  $p_0$ , ambipolar  $D_a$ , and diffusion transit time across  $W = 100 \mu\text{m}$ :  $t \sim W^2/(2D_a)$ .

**Solution.**

*Step 1: Calculate individual diffusion coefficients.*

Using Einstein relation:

$$\begin{aligned}D_n &= \mu_n \frac{kT}{q} = 1200 \times 0.0259 = 31.08 \text{ cm}^2/\text{s} \\ D_p &= \mu_p \frac{kT}{q} = 450 \times 0.0259 = 11.65 \text{ cm}^2/\text{s}.\end{aligned}$$

*Step 2: Determine equilibrium hole concentration.*

$$p_0 = \frac{n_i^2}{n_0} = \frac{(1.0 \times 10^{10})^2}{5.0 \times 10^{15}} = \boxed{2.0 \times 10^4 \text{ cm}^{-3}}.$$

(Minority carriers heavily suppressed due to doping.)

*Step 3: Understand ambipolar diffusion.*

When excess carriers are injected, electrons and holes cannot diffuse independently. The electrostatic coupling (quasi-neutrality) forces them to move together. The effective diffusion coefficient is a weighted average:

$$D_a = \frac{D_n \mu_p p_0 + D_p \mu_n n_0}{\mu_n n_0 + \mu_p p_0}.$$

*Step 4: Calculate denominator (conductivity-like term).*

$$\begin{aligned}\mu_n n_0 + \mu_p p_0 &= 1200 \times 5.0 \times 10^{15} + 450 \times 2.0 \times 10^4 \\ &= 6.0 \times 10^{18} + 9.0 \times 10^6 \approx 6.0 \times 10^{18} \text{ s}^{-1}.\end{aligned}$$



(First term dominates because  $n_0 \gg p_0$ .)

*Step 5: Calculate numerator (diffusivity-weighted term).*

$$\begin{aligned} D_n \mu_p p_0 + D_p \mu_n n_0 &= 31.08 \times 450 \times 2.0 \times 10^4 + 11.65 \times 1200 \times 5.0 \times 10^{15}. \\ &= 2.80 \times 10^8 + 6.99 \times 10^{19} \approx 7.0 \times 10^{19} \text{ cm}^2/\text{s}. \end{aligned}$$

(Second term dominates.)

*Step 6: Calculate ambipolar diffusion coefficient.*

$$D_a = \frac{7.0 \times 10^{19}}{6.0 \times 10^{18}} = \boxed{11.66 \text{ cm}^2/\text{s}}.$$

**Important:**  $D_a \approx D_p$  (minority carrier diffusivity). In doped material, ambipolar diffusion is limited by the minority carrier properties because they are few and create the bottleneck.

*Step 7: Convert thickness and calculate transit time.*

$$W = 100 \mu\text{m} = 1.0 \times 10^{-2} \text{ cm}.$$

For diffusion from the center of a layer, the characteristic diffusion time is:

$$t \sim \frac{W^2}{2D_a} = \frac{(1.0 \times 10^{-2})^2}{2 \times 11.66} = \frac{1.0 \times 10^{-4}}{23.32} = \boxed{4.29 \times 10^{-6} \text{ s}} \approx \boxed{4.3 \mu\text{s}}.$$

**Physical significance:** - Carriers take  $\sim 4 \text{ s}$  to diffuse across  $100 \text{ m}$  - If bulk lifetime is longer than  $4 \text{ s}$ , most carriers are collected before recombination - This is the basis for efficient photodiode design: collection width  $\sim 1\text{--}10 \times L_n$  (diffusion length)

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## 2 Part B — Conceptual Q&A (10 questions)

**Q1. When is the Einstein relation  $D = \mu kT/q$  valid?**

**Answer.** The Einstein relation holds for:

1. **Non-degenerate semiconductors:** Fermi level must be far from band edges (typically  $> 3kT$  away)
2. **Near equilibrium or weak excitation:** When quasi-Fermi level splitting  $\Delta E_F \ll E_g$  (bandgap)
3. **Locally:** Even under non-uniform excitation, the relation applies locally at each point

In extreme high-level injection (quasi-Fermi levels near band edges) or degeneracy, corrections become important.

**Q2. Why do electrons and holes have opposite signs in diffusion current formulas?**

**Answer.** - Electrons diffuse from high to low  $n$ : conventional current is  $J_n = qD_n(dn/dx)$  (opposite to particle flux) - Holes diffuse from high to low  $p$ : conventional current is  $J_p = -qD_p(dp/dx)$  (same as particle flux, but opposite charge) - The minus sign for holes accounts for the fact that hole flux and electron flux are coupled by  $p = -(-e)(\text{hole density})$

### Q3. What is the difference between low-level and high-level injection?

**Answer.**

- **Low-level:**  $\Delta n = \Delta p \ll$  majority carrier concentration. Majority density and mobility unchanged. Minority carrier transport dominates.
- **High-level:**  $\Delta n = \Delta p \gtrsim$  majority carrier concentration. Both bands significantly perturbed. Recombination becomes radiative (GaAs) or Auger (Si) dominated. Mobility may decrease due to increased scattering.

### Q4. What do $n_1$ and $p_1$ represent in the SRH formula?

**Answer.** -  $n_1 = n_i \exp[(E_t - E_i)/kT]$ : electron density that would equalize trap energy with conduction band -  $p_1 = n_i \exp[-(E_t - E_i)/kT]$ : hole density that would equalize trap energy with valence band - They determine the *selectivity* of the trap: whether it captures electrons or holes more readily - **Midgap trap:**  $n_1 = p_1 = n_i$  (perfectly selective for both) - **Shallow trap near  $E_c$ :**  $n_1 \ll n_i$  (captures electrons easily, emits holes slowly)

### Q5. Why are deep traps (near midgap) more effective for SRH recombination than shallow traps?

**Answer.** - Deep traps have  $E_t$  far from both band edges - Both  $n_1$  and  $p_1$  are of order  $n_i$  (neither extremely large nor small) - Both capture channels ( $e^- \rightarrow \text{trap} \rightarrow h^+$ ) are efficient - Shallow traps near  $E_c$ : one channel is suppressed ( $n_1 \ll n_i$  makes electron capture fast, but hole capture blocked) - Result: midgap traps maximize the effective recombination rate

### Q6. When is direct (radiative) recombination the dominant loss mechanism?

**Answer.** - **Direct-gap materials:** GaAs, InP, CdTe (crystal momentum is conserved for band-edge transitions) - **Conditions:** Moderate to high injection ( $10^{16}$ – $10^{18}$  cm $^{-3}$ ), clean material (low defect density), room temperature - **Applications:** LEDs, laser diodes, efficient photodiodes - **Indirect-gap Si:** Radiative lifetime  $\sim 100$ – $1000$  s (phonon assistance needed); always weak compared to SRH

### Q7. How do quasi-Fermi levels relate to the mass action product $np$ ?

**Answer.** Under non-equilibrium, the generalized mass action law is:

$$np = n_i^2 \exp\left(\frac{E_{Fn} - E_{Fp}}{kT}\right),$$

where  $E_{Fn} - E_{Fp} = \Delta E_F$  is the quasi-Fermi level splitting. - At equilibrium:  $\Delta E_F = 0 \Rightarrow np = n_i^2$  - Under excitation:  $\Delta E_F > 0 \Rightarrow np > n_i^2$  (widened splitting drives higher product) - This voltage  $\Delta E_F$  is the *photovoltage* in solar cells and the *threshold voltage* in LEDs

### Q8. What is ambipolar diffusion, and when does it apply?

**Answer.** - When excess electrons and holes are injected together under *quasi-neutrality*, they cannot move independently - Electrostatic coupling forces  $n - p = N_D - N_A$  (locally maintained) - The effective diffusion is a weighted harmonic mean of individual diffusivities:

$$D_a = \frac{D_n \mu_p n_0 + D_p \mu_n p_0}{\mu_n n_0 + \mu_p p_0} \quad (\text{low-level injection}).$$

- In doped material:  $D_a \approx D_{\text{minority}}$  (minority carrier bottleneck) - In intrinsic:  $D_a = (D_n + D_p)/2$  (harmonic mean)

### Q9. Why is quasi-neutrality maintained despite generation/recombination?

**Answer.** The dielectric relaxation time in semiconductors is extremely short:

$$\tau_d = \frac{\varepsilon}{\sigma} \sim 1\text{--}100 \text{ ps.}$$

This is  $\sim 10^6$  times faster than recombination lifetimes ( $\sim \mu\text{s}$ ). - Any local charge imbalance ( $\Delta n \neq \Delta p$ ) is neutralized by current flow in  $\tau_d$  - Over timescales  $\gg \tau_d$ , the bulk remains quasi-neutral:  $n - p \approx N_D - N_A$  - This justifies the use of drift-diffusion equations and Poisson's equation together

### Q10. How does temperature affect recombination and diffusion?

**Answer.**

- **Diffusion:**  $D = \mu \cdot kT/q$ . Mobility  $\mu$  usually decreases with  $T$  (phonon scattering), but  $kT/q$  increases. Net effect:  $D$  often increases slowly with  $T$ .
- **Intrinsic carrier density:**  $n_i \propto T^{3/2} \exp(-E_g/2kT)$ . Increases *rapidly* with  $T$  (exponential term dominates).
- **SRH:** Lifetimes  $\tau_n, \tau_p$  typically decrease with  $T$  (more phonon-assisted emission from traps).
- **Radiative:** Coefficient  $B$  increases with  $T$  (band-edge broadening), but effect is weak.
- **Auger:** Coefficients  $C_n, C_p$  increase with  $T$ . At very high  $T$ , Auger dominates.
- **Net result:** Device performance degrades at high  $T$ : efficiency drops, dark current increases, forward voltage drops.